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Effect of combination therapy of methylfolate with antidepressants in patients with depressive disorder

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Abstract

Objective To determine the relative effectiveness of combination therapy of antidepressants with low-dose methylfolate versus antidepressant monotherapy in patients with depressive disorder.

Methods In an open-label clinical trial, forty-four patients with depressive disorder (6A70, 6A71, and 6A72 according to ICD-11) received an evidence-based antidepressant therapy (either escitalopram 10–20 mg, sertraline 50–100 mg, fluoxetine 20–40 mg, duloxetine 30–60 mg, mirtazapine 15–30 mg, venlafaxine 75–150 mg, trazodone 50–100 mg, amitriptyline 25–75 mg, or clomipramine 25–75 mg orally daily for 4 weeks). The experimental group, Group B was additionally given a dose of methylfolate 800 µg daily for four weeks. The psychometric testing of depression was achieved through administration of Patient Health Questionnaire-9 (PHQ-9) on days 0, 14, and 28, to find the improvement in the depressive score on day 28, the remission rate on day 28, and decrease in the time lag on day 14. Adverse effects were analyzed by self-assessment questionnaire to observe the tolerability. The paired sample t-test and the independent sample t-test were applied for analysis by using SPSS v27.0. The p -value ≤ 0.05 was considered statistically significant.

Results On day 28, patients on low dose methylfolate and antidepressant combination therapy showed a 40.33% improvement in depression symptoms, compared to 26.43% in patients on antidepressant monotherapy (p -value < 0.05). This treatment strategy had no effect on the time-lag of outcomes on day 14 and the remission rate on day 28 was not found statistically significant. Only one patient reported insomnia, while another mentioned irritability.

Conclusion Combination therapy of antidepressants with low-dose methylfolate may provide a safe and effective treatment strategy for patients with depressive disorder.

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Keywords Antidepressants, Depression, Depressive disorders, L-methylfolate, Remission for depression, Adjuvant therapy

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Novelty of the research

What is already known on the subject?

1. It is well-known that the antidepressant medications start improving depressive symptoms after a delay of 2–4 weeks which is identified as the therapeutic latency of antidepressants [1].
2. There is an evidence that pharmacological augmentation can be achieved by using a combination therapy of antidepressants with either bupirone or an antipsychotic agent [2].
3. It is found that resistant depression can be treated by adjunct therapy of methylfolate with SSRIs/SNRIs at a dose of 7.5–15mg/day [3, 4].

What this study has discovered which is new?

1. We showed for the first time that the severity of the depressive symptoms can be improved by the combination therapy of low-dose methylfolate and the antidepressant medication, early in the course of treatment.
2. Furthermore, a combination of low-dose methylfolate and antidepressants is an effective and safe strategy to augment the outcomes of antidepressant therapy.
3. We extended our knowledge of the use of nutraceuticals for the treatment of depression by combining the pharmaceutical and nutraceutical modalities and using methylfolate with antidepressants in our research.

The clinical impact of our work demonstrates that depressive symptoms may be improved early in the course of treatment by the combination therapy of antidepressants and methylfolate as compared to antidepressant monotherapy and it will be a safe treatment strategy.

Introduction

World Health Organization (WHO) statistics depict the 12-month prevalence of depression as 6% while the life-time prevalence varies between 5 and 17% worldwide [2]. Similarly, a rough estimate of the prevalence of depression and chronicity in Pakistan is indicated on the higher side i.e. 13.65% [5]. Many countries are largely affected due to the unavailability of mental health services [6]. Moreover, the lag period of 2–4 weeks [1], poor response to the antidepressant therapy [7], reluctance in follow-up of the treatment and the residual depressive symptoms [8] are still a challenge besides financial incapacity of patients to buy medicines in the vast majority of low-income countries.

In addition to pharmacotherapy, the observational data shows the protective and ameliorative antidepressant

role of some dietary nutrients [9]. One of these nutrients is folate which is available in three main forms: dietary folate, folic acid (FA), and L-methylfolate (MF). FA and dietary folate are converted to MF to function as 1-C transfer. They are essential for the methylation reactions, synthesis of RNA and DNA, and regulation of monoamines (including serotonin) [3].

The folate derivatives have been studied either alone or in combination with selective serotonin reuptake inhibitors for depressive disorders. Out of these, MF has been investigated for depression in a wide range of doses ranging from 400mcg to 15 mg/day [10–12]. It is known that adjunct therapy of high dose MF, 7.5–15 mg/day with selective serotonin reuptake inhibitors (SSRI) or serotonin and norepinephrine reuptake inhibitors (SNRIs) has been approved for the treatment of resistant depression [4, 12–14]. The guidelines were published by the World Federation of Sciences of Biological Psychiatry (WFSBP) and the Canadian Network for Mood and Anxiety Treatments (CANMAT). These authorities have given provisional recommendations for MF as monotherapy and as an adjunct therapy in major depressive disorder (MDD) [15].

Therefore, further studies are required in this domain to document the role of low-dose MF for newly diagnosed cases of depressive disorder, and this clinical trial was designed to determine the clinical outcome of patients taking antidepressant medication in combination with low-dose MF from the beginning which may help many depressive patients at an early stage.

Materials and methods

According to the PICOT (Population, Intervention, Comparison, Outcome, Time) format, the clinical trial strategy was: “In 44 patients with depressive disorder (P), the adjunct therapy of MF (I), compared to antidepressant monotherapy (C), improved the depressive symptoms of the patient (O), over 4 weeks (T)”.

Ethical approval

This study was conducted from 01st Sep’22 to 30th Aug’23 after the approval of the Ethical Review Committee (ERC), of Islamic International Medical College (IIMC), Riphah International University (RIU). The approval number is Riphah/IIMC/IRC/22/2079. It has also been registered to clinicaltrial.gov [16].

Sample size

The sample size (44) was calculated by the WHO sample size formula, with prevalence = 5.9, margin of error = 5, and CI = 95%.

Inclusion criteria

Psychiatric patients with depressive disorder (6A70, 6A71, and 6A72 according to ICD-11), and PHQ-9 score of more than 9 were included in the study [17]. The minimum age considered for the patients was 14 years and there was no upper limit. For Group B (the experimental group), the serum folate levels less than 13.0 ng/mL were included in the study [18].

Exclusion criteria

Psychiatric patients with depressive disorder having a PHQ score less than 9 were excluded [17]. Also, patients with depressive disorders (6A7Y and 6A7Z) and those diagnosed other than depressive disorder according to ICD-11 were not included [19]. We also excluded mentally handicapped and terminally ill patients, and those taking multiple nutritional supplements over the past 6 months.

Data collection

The patients were selected from the sampling frame by convenience sampling technique, presented to the outdoor Psychiatry Department, Pakistan Railway Hospital (PRH), Rawalpindi, Pakistan. They were assessed and diagnosed by a consultant psychiatrist, using ICD-11 criteria and were assigned to either the control group (Group A), who received an antidepressant monotherapy, or the experimental group (Group B), which was prescribed a combination of 800 µg MF daily [10] with antidepressants for 4 weeks. The drugs prescribed during the study were either escitalopram, sertraline, fluoxetine, venlafaxine, duloxetine, amitriptyline, clomipramine or trazodone within the therapeutic range. Data collection was done on days 0, 14 and day 28.

Instrument used

PHQ-9 questionnaire [17].

Improvement in the severity of depressive symptoms at 4 weeks

Total 9 items in the PHQ-9 were rated from 0 to 3, where the sum of scores range was 0–27. The change in PHQ-9 score was recorded.

Lag-time at 2 weeks

The change in the total PHQ-9 score was compared to the baseline score.

Remission at 4 weeks

PHQ-9 scores were divided into class intervals for comprehensive description. Patients with PHQ-9 scores ≤ 4 were labelled as “remission”.

The adverse effects at 4 weeks

The adverse effects (diarrhea, dizziness, insomnia, rash) were assessed by a self-assessment questionnaire, developed by the Department of Psychiatry, PRH, Rawalpindi. The presence or absence of adverse effects due to MF were recorded.

Data analysis

It was done by SPSS v27.0. Descriptive statistics were calculated for the demographic factors and their variability was checked by applying one-way ANOVA at baseline. The mean ($\bar{x}$) and standard deviation (SD) were calculated for PHQ-9 scores on days 0, 14 and 28, and shown as $\bar{x} \pm SD$. The paired sample t-test was applied for within-group analysis (pre-treatment and post-treatment), whereas the independent samples t-test was used to compare the groups. The p -value ≤ 0.05 was considered statistically significant.

Results

Out of a total of 44 patients in this open-label clinical trial, the female-to-male ratio was 4.5:1. The mean and SD of age was 40.34 years $\pm$ 15.13, and this indicates that 68% of the patients had an age between 25.2 and 55.4 years. The frequency and percentages of depression among different age groups are shown in Table 1.

One-way ANOVA was applied to analyze the baseline homogeneity among demographic factors between the study groups. The p -value > 0.05 showed that the treatment groups were similar. Moreover, the box plot showed that the skewness, kurtosis, and variance were within range as checked by SPSS v27.0, and there was an outlier at the baseline reading of group B, as shown in Fig. 1. This is justified by the fact that this patient had borderline PHQ-9 scores, at the time of diagnosis.

Improvement in the severity of depression

The therapeutic benefit in terms of a decrease in PHQ-9 scores and the improvement in the severity of symptoms in percentages on days 14 and 28 are shown in the Table 2.

The paired-sample t-test was applied to analyze pre-post treatment PHQ-9 scores and the differences are shown in Table 3. The p values ≤ 0.05 were considered statistically significant. Table 4 shows a between-group

Table 1 Age distribution of depressive patients

Patients	Frequency	Percent	Valid Percent
Age Groups			
14–24 years	7	15.9	15.9
25–40 years	15	34.1	34.1
41–60 years	15	34.1	34.1
61 years and above	7	15.9	15.9
Total	44	100.0	100.0

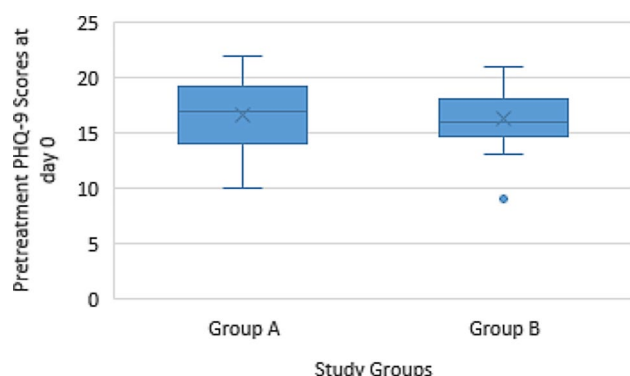


Fig. 1 Box plot showing baseline PHQ-9 scores of Group A and Group B

analysis, indicating better treatment response in group B only on day 28 and not on day 14.

Remission rate on day 28

The data showed a remission rate of 2.27% and 1.14% in patients with antidepressants monotherapy and combination therapy with MF respectively.

Adverse effects in Group B on day 28

One patient in group B, receiving L-MF reported irritability, while another reported insomnia.

Discussion

Depression is rarely diagnosed in the early stage as the symptoms are similar to normal mood swings. The patients only presents themselves to a clinic after realizing that their depression is now really threatening. The late presentation, combined with the associated lag time to antidepressants, contributes to the challenge for the management of moderately severe and severe depressive disorder.

The comprehensive description of the results of this trial, regarding mean and standard deviation showed a decrease in the severity of the depressive symptoms on day 14 and day 28 as measured by PHQ-9. The improvement in PHQ-9 scores was **26.43%** and **40.33%** for Group A and B respectively.

Depression is also termed as the “illness of power and oppression”, which explains that the socially oppressed population especially women in developing countries are more prone and vulnerable to it. All over the world, females suffer from depression more than the male population, which varies from 1.6–3.6:1. Our study showed an F: M ratio of 4.5:1. This female preponderance is universal [20]. We found that depressive disorder was the most prevalent in the adult (25–40 years) and middle age (41–60 years) population.

Table 2 PHQ-9 scores in terms of mean and SD of each study group on day 0,14, and 28

Groups	PHQ-9 score at baseline (Mean ± SD)	PHQ-9 at day 14 (mean ± SD)	% Improvement	PHQ-9 at day 28 (mean ± SD)	% Improvement
Group A	16.68 ± 3.41	14.54 ± 4.26	12.82%	12.27 ± 5.40	26.43%
Group B	16.23 ± 2.58	13.22 ± 3.33	18.49%	9.81 ± 4.22	40.33%

Table 3 Treatment response differences in study groups A&B, at day 28, by applying paired sample t-test

		Paired Differences				t	Sig. (2-tailed)
		Group Means	Mean Diff.	SD	SEM		
Pair 1	Baseline PHQ-9 scores of Group A - PHQ-9 scores at day 28 of Group A	16.68–12.27	4.41	3.22	0.69	6.429	<0.001***
Pair 2	Baseline PHQ-9 scores of Group B - PHQ-9 scores at day 28 of Group B	16.23–9.82	5.41	4.23	0.90	6.001	<0.001***

The *p* value was considered significant at the *0.05, **0.01 and ***0.001

Table 4 Independent sample t test comparing Group A and B

		Levene's Test for Equality of Variances						95% Confidence Interval of the Difference	
		F	Sig.	T	Sig. (2-tailed)	Mean Difference	Std. Error Difference	Lower	Upper
Pre	Equal variances assumed	3.836	0.057	0.498	0.621	0.455	0.912	-1.387	2.296
	Equal variances not assumed			0.498	0.621	0.455	0.912	-1.391	2.300
Mid	Equal variances assumed	2.546	0.118	1.142	0.260	1.318	1.154	-1.011	3.647
	Equal variances not assumed			1.142	0.260	1.318	1.154	-1.015	3.651
Post	Equal variances assumed	15.475	0.001*	1.998	0.052*	2.591	1.297	-0.026	5.208
	Equal variances not assumed			1.998	0.054*	2.591	1.297	-0.052	5.234

The groups had no statistically significant difference regarding demographic factors such as gender, age, occupation, or geographical distribution, that is, $p > 0.05$, showing that all the groups were similar and there were no confounders.

Antidepressant medication is very important treatment for the management of depressive disorders. We have included antidepressants in their therapeutic doses within each study group since all approved medications in this class modulate mood and behavior [21]. In addition, not a single drug suits all patients, and if any differences are reported in their efficacy, that are attributed to physicians' wrong prescription patterns [22].

Group A, the control group [23] showed 12.82% and 26.43% improvement in PHQ-9 scores on days 14 & 28, respectively. Our findings are consistent with the fact that the decline of PHQ-9 scores ≥ 6 indicates a significant improvement in depression at 4 weeks, corresponding to a 20% response [24, 25].

Group B, showed 18.49% and 40.33% improvement after 2 and 4 weeks of treatment. This improvement is statistically significant ($p\text{-value} \leq 0.05$) on day 28 when compared to antidepressant monotherapy.

This has been reflected in a study that used MF without antidepressant medication [11]. Moreover, a number of studies in the past had used it in high doses (7.5 mg–15 mg/day) with resistant depression [26]. So, this trial will give a novel contribution to the clinical practice if supported by more experimental studies. Early supplementation at the time of diagnosis is also important because we already know that the depressive patients delay their presentation to psychiatrist.

Our clinical trial showed that this combination augmented effects on day 28. The remission and time lag improved in terms of frequency and percentages. Non-compliance with the patients is a major dilemma that leads to withdrawal of medication. One of its reason is an initial delay in the improvement of mood and interest due to the lag period. Jakobson et al. [7], debated that the side effects of antidepressant therapy need to be considered, in this regard. The results of our study showed that these supplements are relatively safe. Only one of the patients in group B reported insomnia and another complained irritability.

Integrative medicine (IM) and Complementary and Alternative Medicine (CAM) can help pharmaceutical trends, by utilizing the benefits of nutraceuticals [27]. The use of supplements becomes more important in lower-middle income countries like Pakistan where a large population is already malnourished [6].

The augmentation of antidepressants by adding supplements like methylfolate may help [1] early improvement, and [2] may prevent the adverse effects. So, we can take advantage by combining these two approaches, i.e.

evidence-based pharmacological therapy with supplementation strategies for a better outcome [28–30].

Conclusion

This research aimed to find the benefits of taking dietary supplements (inj/tab), in combination with antidepressants for the treatment of depression. Prescribing dietary supplements, with antidepressants may result in rapid/early improvement of depression, by increasing their efficacy. To avoid any adverse effects, the dose is kept minimum, complemented with prior blood tests according to the specific supplements. Moreover, as per the ethical guidelines, the possible adverse effects of the supplements were explained to the patient. The patient also had the right to voluntarily withdraw from the study, at any point, without any negative consequences, while their rightful entitlements were protected during the trial.

Results

It has been found that the combination therapy of antidepressant medication together with MF (800mcg dose) improved the depressive symptoms of patients. Augmentation of antidepressant therapy, in combination with MF is found to be a safe strategy to enhance and accelerate the response to treatment.

Limitations

An apparent limitation of our study is that this one is a single centered study. Randomization and blinding were not done to ensure the safety and compliance of the patients. In the future, we aim to conduct an extended study to observe the change in remission and lag time as compared to antidepressant monotherapy.

Recommendations

Folate deficiency is implicated as a risk factor for MDD and is also associated with greater severity of depressive symptoms and poor responsiveness to antidepressants. While supplementation with folic acid has been shown to improve depressive symptoms, polymorphisms in the gene which encode the MTHFR enzyme may reduce some patients' ability to convert dietary folate and folic acid into active L-methylfolate. Supplementation with active L-methylfolate therefore may be efficacious in treating depressive symptoms.

Abbreviations

ANOVA	Analysis of Variance
CAM	Complementary and Alternative Medicine
CANMAT	Canadian Network for Mood and Anxiety Treatments
CI	Confidence Interval
DNA	Deoxyribonucleic Acid
ERC	Ethical Review Committee
FA	Folic Acid
IIMC	Islamic International Medical College
ICD	International Classification of Diseases
IM	Integrative medicine

IRC	Institutional Review Committee
MDD	Major depressive disorder
MF	Methylfolate
PHQ	Patient Health Questionnaire
PICOT	Population, Intervention, Comparison, Outcome, Time
RIU	Riphah International University
RNA	Ribonucleic Acid
SD	Standard Deviation
SPSS	Statistical Package for Social Sciences
WFSBP	World Federation of Sciences of Biological Psychiatry
WHO	World Health Organization

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Author contributions

The authors confirm contribution to the paper as follows: study conception and design: AS, MM, UN; data collection: AS, MM, AW; analysis and interpretation of results: AS, SA, MI; draft manuscript preparation: AS, UN, MM. All authors reviewed the results and approved the final version of the manuscript.

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Data availability

Data is provided within the manuscript.

Declarations

Statement of compliance with the declaration of Helsinki

This clinical trial was conducted in full compliance with the ethical principles outlined in the Declaration of Helsinki, as adopted by the World Medical Association (WMA). These principles are dedicated to safeguarding the rights, well-being, and safety of all research participants. All aspects of this study, including its design, implementation, and reporting, will adhere to the latest revision of the Declaration of Helsinki. This includes the requirement for informed consent, ensuring that participants are fully informed about the study's purpose, procedures, risks, and benefits. Special attention will be given to protecting vulnerable populations, and measures will be taken to minimize potential harm while maximizing the anticipated benefits of the research. This study also complies with all relevant national and local regulations, as well as institutional policies governing research ethics. The study protocol has been reviewed and approved by an independent ethics committee or institutional review committee (IRC).

Ethics approval and consent to participate

The study has been approved by the Institutional Review committee (IRC), Islamic International Medical College (IIMC), Riphah International University. (Ref: Riphah/IIMC/IRC/22/2079).

Institutional/Ethical review committee members

Prof. Dr. Maqsood ul Hassan, Prof. Dr. Muhammad Ayaz Bhatti, Prof. Dr. Shabana Ali, Prof. Dr. Shazia Qayyum, and Prof. Dr. Amena Rahim.

Consent to participate

All participants in the clinical trial agreed to take part in this research. They were ensured that their name will not be disclosed anywhere for confidentiality, although their medical record can be used in future research.

Consent for publication

The authors give their consent for the publication of identifiable details to be published in the journal and article.

Competing interests

The authors declare no competing interests.

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